

Phase 1: Brainstorming & Ideation

OptiCrop: Smart Agricultural Production Optimization

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Project Title

OptiCrop: Smart Agricultural Production Optimization Using ML

Problem Statement

Traditional crop selection relies heavily on empirical knowledge and guesswork, often leading to suboptimal yield, wasted resources, and soil degradation. Farmers struggle to accurately correlate complex environmental variables (like NPK ratios, temperature, and humidity) with the specific crop that will yield the best results. Without data-driven insights, there is a high risk of crop failure and financial loss.

Proposed Solution

OptiCrop resolves this by leveraging Machine Learning to analyze soil and climatic conditions. We developed a predictive model using Logistic Regression, trained on historical agricultural data. This model is deployed via a user-friendly, three-page Flask web application. Farmers simply input their land conditions, and the system instantly recommends the most suitable crop, ensuring maximum yield and resource efficiency.

Target Audience

- **Primary:** Farmers and agricultural workers seeking data-driven crop recommendations.
- **Secondary:** Agricultural researchers, students, and policy makers looking to understand the interplay between environmental factors and crop growth.

Technology Stack

- **Programming Language:** Python 3.8+
- **ML Libraries:** Scikit-Learn (Logistic Regression, K-Means, Decision Tree, Random Forest)
- **Data Analysis:** Pandas, NumPy, Seaborn, Matplotlib
- **Web Framework:** Flask
- **Serialization:** Pickle